

A Proof of Conjecture 00000000049

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Claude (Opus 5)[†]

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Abstract

Conjecture 00000000049 asserts, of point sets in the plane all of whose pairwise distances are prime, that the maximal size is 4 and that a set of size 4 exists. Both hold. A set of size 4 is $\{0, 2, 5, 7\}$ on a line, with distances 2, 5, 7, 3, 5, 2. For the bound we show that any four such points are *collinear*, whence five would be collinear too, which a parity argument on the gaps forbids.

1 Statement and conventions

We identify the Euclidean plane with $\mathbb{C}$. Call a finite set $S \subset \mathbb{C}$ a *prime-distance set* if $|P - Q|$ is a prime integer for all distinct $P, Q \in S$. Conjecture 00000000049 asserts

- (a) every prime-distance set has at most 4 elements;
- (b) some prime-distance set has 4 elements.

Throughout, “prime” means a rational prime $2, 3, 5, 7, \dots$; note that 2 is allowed, and it is the only even one. This is what makes the problem differ from the classical question about odd integral distances.

2 Part (b)

Proposition 1. $S = \{0, 2, 5, 7\} \subset \mathbb{R} \subset \mathbb{C}$ is a prime-distance set of size 4.

Proof. The six distances are

$$|0 - 2| = 2, \quad |0 - 5| = 5, \quad |0 - 7| = 7, \quad |2 - 5| = 3, \quad |2 - 7| = 5, \quad |5 - 7| = 2,$$

all prime. □

3 Part (a)

3.1 A determinant identity

Let P_0, P_1, P_2, P_3 be four points of the plane, write $v_i = P_i - P_0$, and put

$$a_i = |v_i| = |P_i P_0|, \quad c_{ij} = |v_i - v_j| = |P_i P_j|.$$

The Gram matrix $G = (v_i \cdot v_j)_{1 \leq i, j \leq 3}$ is singular, because v_1, v_2, v_3 lie in a 2-dimensional space. Since $2 v_i \cdot v_j = a_i^2 + a_j^2 - c_{ij}^2$ for $i \neq j$ and $2 v_i \cdot v_i = 2a_i^2$, setting

$$x_{ij} = a_i^2 + a_j^2 - c_{ij}^2$$

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and expanding $\det(2G) = 0$ gives, after dividing by 2,

$$E := 4a_1^2a_2^2a_3^2 + x_{12}x_{13}x_{23} - a_1^2x_{23}^2 - a_2^2x_{13}^2 - a_3^2x_{12}^2 = 0. \quad (1)$$

We use (1) modulo 8. Recall that an odd square is $\equiv 1 \pmod{8}$, while $2^2 = 4$.

3.2 The distance-two graph

Let S be a prime-distance set and let H be the graph on S whose edges are the pairs at distance exactly 2.

Lemma 2. *Every vertex of H has degree at most 2.*

Proof. Suppose B, C, D all lie at distance 2 from A . They are distinct points of the circle of radius 2 about A , so their pairwise distances are at most the diameter 4 and are prime, hence lie in $\{2, 3\}$. A triangle with sides a, b, c has circumradius $R = abc/(4K)$ with $16K^2 = 2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4$, so $R = 2$ would require $(abc)^2 = 4 \cdot 16K^2$. For the four possibilities this reads

$$(2, 2, 2) : 64 \neq 192, \quad (2, 2, 3) : 144 \neq 4 \cdot 63, \quad (2, 3, 3) : 324 \neq 4 \cdot 128, \quad (3, 3, 3) : 729 \neq 4 \cdot 243,$$

so the circumradius is $2/\sqrt{3}, 4/\sqrt{7}, 9\sqrt{2}/8, \sqrt{3}$ respectively, never 2. $\square$

Lemma 3. *H contains no 4-cycle.*

Proof. Suppose $|AB| = |BC| = |CD| = |DA| = 2$. Both B and D lie on the perpendicular bisector of AC ; writing $e = |AC|$ and M for the midpoint of AC we get $|MB| = |MD| = \sqrt{4 - e^2/4}$, so B and D are symmetric about M and $f = |BD| = 2\sqrt{4 - e^2/4}$, that is

$$e^2 + f^2 = 16.$$

No two primes satisfy this: $2^2 + 2^2 = 8$, $2^2 + 3^2 = 13$, $3^2 + 3^2 = 18$, and any prime ≥ 5 already has square > 16 . $\square$

By Lemmas 2 and 3, on four points H is a disjoint union of paths and triangles. Up to isomorphism the possibilities are: no edge, one edge, two adjacent edges, two disjoint edges, a path P_4 , and a triangle plus an isolated vertex.

3.3 The cases settled by (1)

Lemma 4. *A prime-distance set of size 4 has at least two edges in H , and they are not adjacent.*

Proof. In each case choose the base point P_0 as indicated and reduce (1) modulo 8, using $a^2 \equiv 1$ for odd a and $2^2 = 4$.

No edge (all six distances odd). Then $x_{ij} \equiv 1 + 1 - 1 = 1$ and $x_{ij}^2 \equiv 1$, so

$$E \equiv 4 + 1 - 1 - 1 - 1 = 2 \pmod{8}.$$

One edge. Take P_0 outside that edge, so the edge is $c_{12} = 2$. Then $x_{12} \equiv 1 + 1 - 4 \equiv 6$, $x_{12}^2 \equiv 4$, while $x_{13}, x_{23} \equiv 1$, so

$$E \equiv 4 + 6 - 1 - 1 - 4 = 4 \pmod{8}.$$

Two adjacent edges. Say $|AB| = |AC| = 2$; take $P_0 = D$, so $c_{12} = c_{13} = 2$. Then $x_{12}, x_{13} \equiv 6$ with squares $\equiv 4$, and $x_{23} \equiv 1$, so

$$E \equiv 4 + 36 - 1 - 4 - 4 \equiv 4 + 4 - 1 - 4 - 4 = -1 \equiv 7 \pmod{8}.$$

In all three cases $E \neq 0$, contradicting (1). $\square$

3.4 The remaining configurations

Lemma 5. *H is not a path on four vertices.*

Proof. Suppose $|AB| = |BC| = |CD| = 2$ and these are the only edges of H . Since A and C both lie at distance 2 from B , $|AC| \leq 4$ is prime and is not 2, so $|AC| = 3$; likewise $|BD| = 3$. Place $B = (0, 0)$, $C = (2, 0)$. From $|AB| = 2$, $|AC| = 3$ we get $A = (-\frac{1}{4}, \pm\frac{\sqrt{63}}{4})$, and from $|CD| = 2$, $|BD| = 3$ we get $D = (\frac{9}{4}, \pm\frac{\sqrt{63}}{4})$. Hence

$$|AD|^2 = \left(\frac{10}{4}\right)^2 + \left(\pm\frac{\sqrt{63}}{4} \mp \frac{\sqrt{63}}{4}\right)^2 = \frac{25}{4} \quad \text{or} \quad \frac{25}{4} + \frac{63}{4} = 22,$$

so $|AD| = \frac{5}{2}$ or $\sqrt{22}$, neither an integer. $\square$

Lemma 6. *H is not a triangle together with an isolated vertex.*

Proof. Let A, B, C be equilateral of side 2 and let D have odd prime distances $p = |DA|$, $q = |DB|$, $r = |DC|$. Place $A = (0, 0)$, $B = (2, 0)$, $C = (1, \sqrt{3})$ and $D = (x, y)$. From $|DA|, |DB|$,

$$x = \frac{p^2 - q^2 + 4}{4},$$

and $p^2 - q^2 \equiv 0 \pmod{8}$, so x is an odd integer; then $y^2 = p^2 - x^2 \equiv 1 - 1 = 0 \pmod{8}$. The third distance gives $p^2 - 2x + 4 - r^2 = 2\sqrt{3}y$, hence

$$N^2 = 12y^2, \quad N := p^2 - 2x + 4 - r^2.$$

Now $p^2 - r^2 \equiv 0 \pmod{8}$ and $2x \equiv 2$ or $6 \pmod{8}$, so $N \equiv 2$ or $6 \pmod{8}$; in particular $N \equiv 2 \pmod{4}$, say $N = 2m$ with m odd, whence $N^2 = 4m^2 \equiv 4 \pmod{32}$. On the other side $8 \mid y^2$ gives $12y^2 \equiv 0 \pmod{32}$. So $4 \equiv 0 \pmod{32}$, absurd. $\square$

Lemma 7. *If H consists of two disjoint edges, the four points are collinear.*

Proof. Say $|AB| = 2 = |CD|$ and the other four distances are odd primes. Place $A = (-1, 0)$, $B = (1, 0)$. For any point $P = (x, y)$ with $|AP|, |BP|$ odd,

$$x = \frac{|AP|^2 - |BP|^2}{4}$$

is an even integer, because $|AP|^2 - |BP|^2 \equiv 0 \pmod{8}$; and $y^2 = |AP|^2 - (x+1)^2 \equiv 1 - 1 = 0 \pmod{8}$. Apply this to $C = (x_C, y_C)$ and $D = (x_D, y_D)$.

From $|CD|^2 = 4$ and $x_C - x_D$ even we get $(x_C - x_D)^2 \in \{0, 4\}$.

If $x_C = x_D$ then $(y_C - y_D)^2 = 4$, so $2y_C y_D = y_C^2 + y_D^2 - 4 \equiv 4 \pmod{8}$; thus $y_C y_D$ is an integer $\equiv 2 \pmod{4}$ and $(y_C y_D)^2 \equiv 4 \pmod{32}$. But $(y_C y_D)^2 = y_C^2 y_D^2 \equiv 0 \pmod{64}$. Contradiction.

So $|x_C - x_D| = 2$ and $y_C = y_D =: y$. Relabelling, write $C = (x, y)$, $D = (x+2, y)$ and put $p = |AC|$, $q = |AD|$, $r = |BC|$; then $|BD| = |AC| = p$ and

$$q^2 + r^2 = 2p^2 + 8, \quad x = \frac{p^2 - r^2}{4}, \quad y^2 = p^2 - (x+1)^2.$$

Now $y^2 \geq 0$ forces $(p-2)^2 \leq r^2 \leq (p+2)^2$, i.e. $p-2 \leq r \leq p+2$; as p, r are odd, $r \in \{p-2, p, p+2\}$. If $r = p$ then $q^2 = p^2 + 8$, so $(q-p)(q+p) = 8$ with both factors even, forcing $q-p = 2$, $q+p = 4$ and $p = 1$, not prime. Hence $r = p \pm 2$, which gives $x+1 = \pm p$ and $y^2 = 0$: the four points lie on the x -axis. $\square$

Theorem 8. *Every prime-distance set of size 4 is collinear.*

Proof. By Lemma 4 the graph H has at least two, pairwise non-adjacent, edges; with Lemmas 2 and 3 the only remaining possibilities on four vertices are two disjoint edges, a path P_4 , or a triangle plus an isolated vertex — and the last two are excluded by Lemmas 5 and 6. So H is two disjoint edges, and Lemma 7 applies. $\square$

3.5 No five points

Lemma 9. *No five collinear points form a prime-distance set.*

Proof. Let the points be $x_0 < \dots < x_4$ with consecutive gaps $g_i = x_i - x_{i-1}$, $i = 1, \dots, 4$. Every g_i is prime, and so is every sum of consecutive gaps. Two adjacent odd gaps would give an even sum ≥ 6 , not prime; two adjacent gaps equal to 2 would give the sum 4, not prime. So of each adjacent pair exactly one equals 2, and the pattern alternates:

$$(2, \text{odd}, 2, \text{odd}) \quad \text{or} \quad (\text{odd}, 2, \text{odd}, 2).$$

In both cases the total span $g_1 + g_2 + g_3 + g_4$ is 4 plus two odd primes, hence even and at least 10 — not prime. But the span is $x_4 - x_0$, which must be prime. $\square$

Theorem 10. *Every prime-distance set has at most 4 elements.*

Proof. Suppose S is a prime-distance set with $|S| \geq 5$ and pick five points of it. Every four of them form a prime-distance set, hence are collinear by Theorem 8. Fix four of them; they lie on a line L . Any fifth point P , together with three of those four, forms a collinear quadruple; as three distinct points of L determine L , we get $P \in L$. So all five lie on L , contradicting Lemma 9. $\square$

Together with Proposition 1, Conjecture 00000000049 holds.

4 Computational corroboration

Four points are planar exactly when their 5×5 Cayley–Menger determinant vanishes; with prime squared distances this is an exact integer test. Over all 6-tuples of primes below 120 satisfying the triangle inequalities — about $4.4 \cdot 10^6$ configurations — the only solutions are degenerate, and every one of them is $\{0, 2, 5, 7\}$ up to relabelling and reflection. This agrees with Theorem 8. The search is corroboration, not part of the proof.

5 Formal verification

The accompanying Lean 4 project formalizes, against Mathlib:

- `IsPrimeDistanceSet`, `ConjectureHolds` — the notion and the conjecture, both parts, with the plane modeled as $\mathbb{C}$;
- `witness`, `dTable_prime`, `conjecture_part_b` — Proposition 1, that is **part (b) in full**;
- `not_prime_add_of_ne_two`, `eq_two_of_add_prime`, `no_five_collinear` — Lemma 9 in full, the last step of part (a);
- `no_prime_sq_add_sq_eq_sixteen` (Lemma 3), `no_prime_sq_eq_sq_add_eight` (the $r = p$ branch of Lemma 7), `circumradius_ne_two` (Lemma 2) — the arithmetic obstructions on which the case analysis turns.

Scope. Part (b) is formalized completely. Part (a) is *not*: the Gram identity (1) and its reductions modulo 8 (Lemma 4), the coordinate computations of Lemmas 5, 6 and 7, and the passage from Theorem 8 to the five-point bound are carried out in this paper only. What the Lean development contributes to part (a) is its final collinear step together with the three arithmetic facts the case analysis appeals to. We record this explicitly rather than leave the coverage to

be inferred; a full formalization would require developing the plane-geometric apparatus above in Lean and is a substantially larger undertaking than the present one.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with `Mathlib v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_part_b` and `#print axioms no_five_collinear` report exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.